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LABORATORY-SCALE PLANETARY SURFACE MAPPING USING USRP-BASED SYNTHETIC APERTURE RADAR

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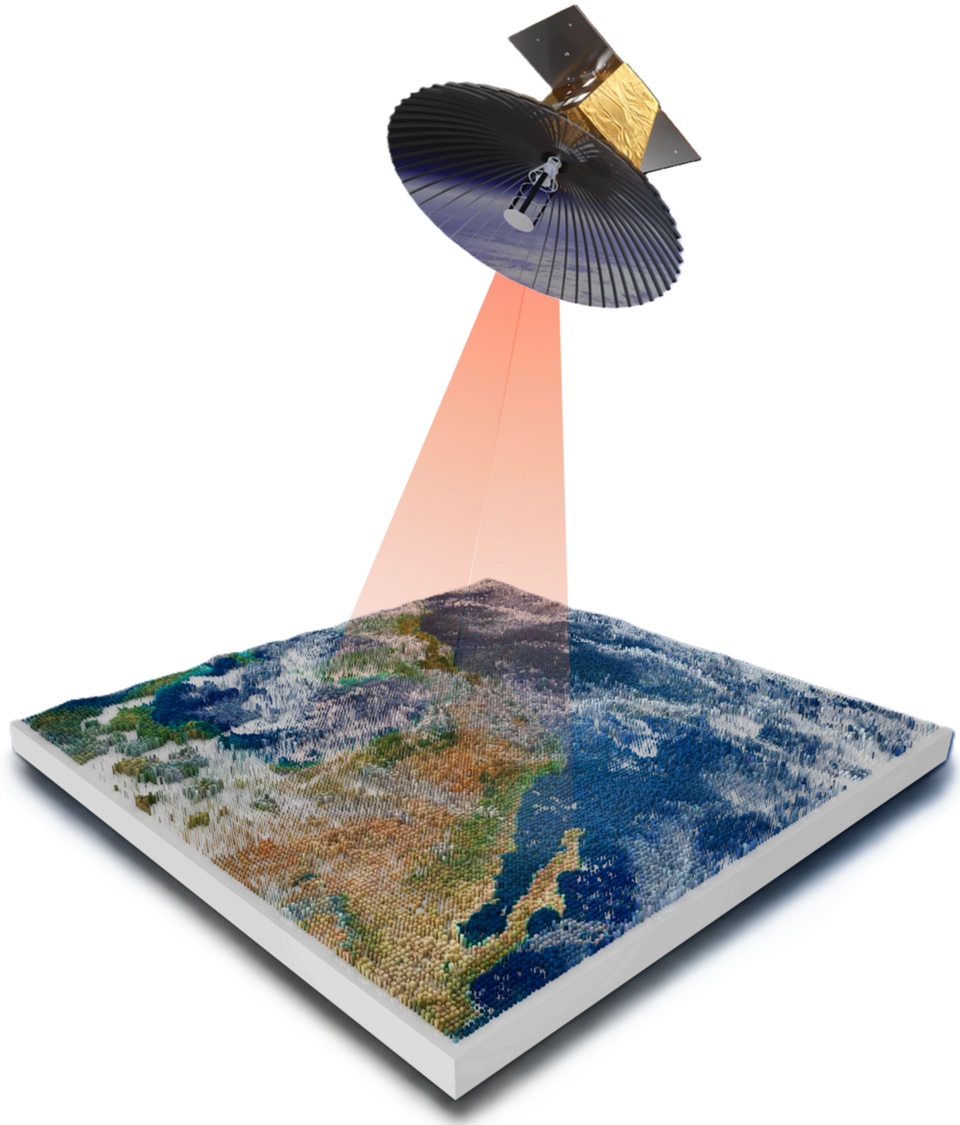
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INTRODUCTION

- **Planetary surface mapping** is essential for understanding the geology, morphology, and physical properties of planets and moons.
- Planetary radar mapping serves as a critical bridge between standard photography and physical landers. Because **RADAR is an active sensor** meaning it provides its own "light" in the form of radio waves it can reveal data that **optical cameras simply cannot see.**



WHY RADAR ?

- Radar-based remote sensing is widely used in planetary exploration because it **can operate in darkness and penetrate clouds, dust, and surface layers** where optical systems fail.
- Microwave radar systems transmit electromagnetic waves toward a surface and analyze the reflected echoes to **extract information such as range, reflectivity, and surface roughness.**
- **Synthetic Aperture Radar (SAR)** improves spatial resolution by coherently combining radar measurements collected from multiple antenna positions.

PROBLEM STATEMENT

Conventional planetary surface mapping techniques **primarily rely on optical and imaging sensors**, which are highly dependent on sunlight and are **significantly affected by atmospheric conditions**. These methods provide limited information about surface roughness and material properties and are ineffective for subsurface or all-weather observation. RADAR based techniques, particularly **Synthetic Aperture RADAR (SAR)**, overcome these limitations by enabling high-resolution surface characterization **independent of illumination and environmental conditions**. However, existing RADAR and SAR systems used in **planetary missions rely on highly specialized and costly instrumentation**, which limits their practical implementation at smaller scales. This creates the need for a **low-cost, flexible platform such as the Universal Software Radio Peripheral (USRP)** to demonstrate radar-based planetary surface mapping concepts.

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